On The Gravitational Nature Of Time

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Abstract

We investigate whether the Lorentz factor can be represented as the slope of an additional spatial coordinate $\boldsymbol{y}$, $\boldsymbol{\gamma = \sqrt{1 + (\partial_x y)^2}}$, in a Kaluza–Klein-like construction in which the extra coordinate is not compactified. The coordinate transformations preserve the Minkowski interval for instantaneous events. Treating the gradient of $\boldsymbol{y}$ as a pure-gauge four-potential, the induced radial metric reproduces the spatial geometry of the Schwarzschild solution as a Flamm-type embedding, and the metric component $\boldsymbol{G_{44}}$ is fixed to unity, consistent with the Cassini bound on the post-Newtonian parameter $\boldsymbol{\gamma}$. The same relation, $\boldsymbol{\beta^2 = 2Gm(R)/(c^2 R)}$, describes the flow velocity in the Painlev'e–Gullstrand form of Schwarzschild geometry and in the Hubble flow of a flat universe, with opposite sign. Where the construction can be checked it is equivalent to general relativity, and the temporal metric function is supplied by a postulate rather than derived. We outline the field equations, the non-gradient potential and the dimensional treatment needed for the construction to make predictions that differ from general relativity.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of the geometry of SR under the assumption that the $\mathbb{R}^3$ space between two arbitrary points elongates as time progresses.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, y, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6]. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process

as observed from different reference frames in a space-time geometry where the 4th spatial KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained by attributing the additional spatial separation to the 4th spatial dimension, $dy = v\tau\sqrt{\gamma^2 - 1}$ as shown in figure 1. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th spatial axis.

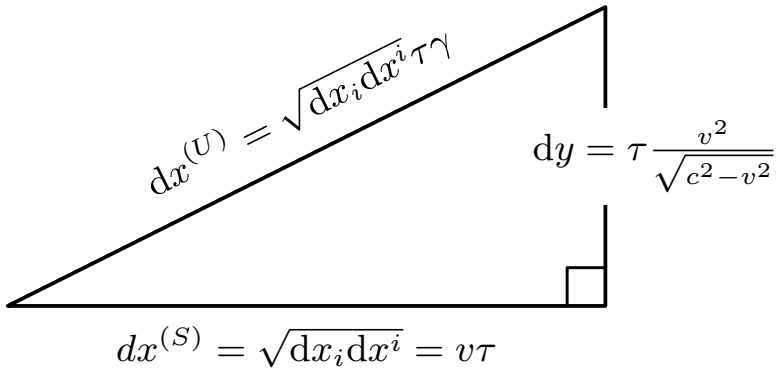


Fig. 1 Right-triangle construction used to represent the ordinary spatial displacement dx, the additional spatial displacement dy, and the extended displacement $dx^{(U)}$. The relation assigned to the hypotenuse is an assumption of the proposed geometric construction.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event represents a local measurement as $dt \to 0$.

Let there be two overlapping $\mathbb{R}^3$ coordinate systems, **S** and **U**. Let **S** and **U** represent an equi-temporal surface within a 4-dimensional hyperspace where $y = y^{(S)}$ and $y = y^{(U)}$ in each respective 3-dimensional surface.

This gives

$$f\left(x_i, y\right) = \partial_y \mathcal{H} \tag{1}$$

where

$$f\left(x_i, y^{(A)}\right) = \mathbf{A}\left(x_i\right) \tag{2}$$

for an arbitrary **A** contained within $\mathcal{H}$.

Let **U** represent a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{3}$$

where $\Delta t > 0$ and $\mu = 1, 2, 3, 4$. Likewise, let $\mathbf{S}$ and $\mathbf{U}$ be bound such that

$$f\left(x_i, y^{(U)}\right) = f\left(x_i, y^{(S)}\right) + \int_{x_0^{(S)}}^{x_0^{(U)}} F(x)\ dt \tag{4}$$

where F is the sum of all forces acting on the point of space at x, integrated according to time, which in turn sums the change between the temporal surfaces $\mathbf{S}$ and $\mathbf{U}$.

Let there also be two observers, A and B. Let A and B agree that B should travel at some *agreed upon* relativistic velocity v' between two arbitrary points in space contained within $\mathcal{H}$, p_1 and p_2. Let the experiment be arranged such that B passes p_1 in $\mathbf{S}$, and p_2 in $\mathbf{U}$.

Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

We then make these assertions

1. To maintain a classical model of simultaneity A and B must agree upon the time at which B reaches p_2.
2. To satisfy existing experimental evidence for the additional γ factor[7–11], $\Delta t = \gamma \frac{d}{v}$.

This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \frac{\overbrace{d \cdot \gamma}^{B}}{v^{(U)}} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{5}$$

and as v' dilates inversely with coordinate density, $v'^{(S)} = v'^{(U)}/\gamma$.

2 Theoretical Framework

We acknowledge two primary properties of time:

1. A driving rate of change, at least as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance beyond the two properties described above. We presume that time acts only as a mathematical tool: $\Delta t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart with only the relativistic portion of associated magnitudes protruding out of the containing temporal surface.

To remove time as a physically meaningful quantity, we let

$$\tau = \frac{\dot{x}}{\widehat{x}} \tag{6}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change, a magnitude of length that grows with an arbitrary temporal scalar. Let $\widehat{x} = \partial_\tau x$ $\hat{\tau} = |\vec{x}|$ where $\hat{\tau} = 1$, or the velocity in units of distance. τ, while equivalent in magnitude to t is derived directly from the geometry and has no explicit time dependence.

2.1 Dictionary

Table 1 Quantities of this paper and their standard counterparts

This paper	Standard counterpart
$\beta c = v$	shift vector ("river velocity"), escape velocity from the enclosed mass
γ	Lorentz factor between the flow and static observers; $1/\gamma$ is the static lapse
$y(R)$, $\partial_R y = \beta\gamma$	Flamm embedding of the static slice, $d\ell^2 = dR^2 + dy^2$
$\partial_\tau y = v^2/\sqrt{c^2 - v^2}$	dy/dT along a free-fall worldline
$\delta(M, R) = 2GM/R^3$	radial tidal eigenvalue
v_{orbit}	βc, the escape velocity of the enclosed mass
v_{cosmo}	$c\,(v_{\text{esc}}/R)\,\tau$, with v_{esc}/R the Hubble-like rate of the flow at R
$v_{\text{kinematic}}$	$c\,\Omega\,\tau$ with $\Omega = \sqrt{GM/R^3}$ the circular-orbit angular frequency

The last two rows follow from $v = c\sqrt{1 - \varphi^{-2}} \approx c\sqrt{2(\varphi - 1)}$ with $\varphi - 1 = GM/R^3$ and $\frac{1}{2}GM/R^3$ respectively, and they explain why the two velocities differ by a factor $\sqrt{2}$, the ratio of escape to circular speed.

Remark. The same factor applies to equation 49. The velocity obtained there is βc, the escape velocity of the enclosed mass. In Schwarzschild geometry the circular velocity is $\beta c/\sqrt{2}$ to leading order. This is the origin of the discrepancy noted earlier between $v \approx v_{\text{escape}}$ and a stable orbit, and a derivation of the inertial correction is needed before equation 49 can be read as a circular velocity.

2.2 Lorentz Invariance

We consider these coordinate transformations with regards to the Lorentz transformations and find that for an instantaneous event where $dt \to 0$

$$\boxed{\begin{aligned} dx' &= \gamma dx = v^{(U)} \\ c\,dt' &= -\gamma\frac{v}{c}dx = -\partial_\tau y \end{aligned}} \tag{7}$$

where y is as defined in equation 9.

We define $v\,(\gamma)$ to be the Lorentz factor solved for v

$$v\,(\gamma) = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{8}$$

Beginning with the definition of the additional coordinate,

$$y = \tau \frac{v^2}{\sqrt{c^2 - v^2}} \tag{9}$$

we take the partial derivative with respect to τ, holding v constant:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{10}$$

Since $x = v\tau$, we can equivalently express the coordinate as a function of x:

$$y = x \frac{v}{\sqrt{c^2 - v^2}}. \tag{11}$$

Taking the partial derivative with respect to x gives

$$\boxed{\partial_x y = \frac{v}{\sqrt{c^2 - v^2}}} \tag{12}$$

We can now express this derivative in terms of the Lorentz factor:

$$\partial_x y = \frac{\gamma v}{c}. \tag{13}$$

Multiplying by the velocity expressed as a function of γ,

$$\begin{aligned} v(\gamma)\partial_x y &= c\sqrt{1 - \frac{1}{\gamma^2}}\frac{\gamma v}{c} \\ &= v\sqrt{\gamma^2 - 1} \\ &= \frac{\gamma v^2}{c}. \end{aligned} \tag{14}$$

Finally we recover

$$\boxed{v(\gamma)\partial_x y = \frac{\gamma v^2}{c} = \partial_\tau y} \tag{15}$$

2.2.1 Invariance of the Minkowski Interval

We can further examine the Lorentz invariance of these transformations by considering the Minkowski interval. For an instantaneous event, where $dt \to 0$, the transformed interval is given by

$$ds'^2 = c^2 dt'^2 - dx'^2. \tag{16}$$

Substituting the coordinate transformations established above,

$$
\begin{aligned}
ds'^2 &= \left(-\gamma \frac{v}{c} dx\right)^2 - (\gamma dx)^2 \\
&= \gamma^2 dx^2 \left(\frac{v^2}{c^2} - 1\right).
\end{aligned}
\tag{17}
$$

Using the definition of the Lorentz factor,

$$
\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}},
\tag{18}
$$

we obtain

$$
ds'^2 = -\gamma^2 dx^2 \left(1 - \frac{v^2}{c^2}\right).
\tag{19}
$$

Therefore,

$$
\boxed{ds'^2 = -dx^2 = ds^2}
\tag{20}
$$

Thus, the instantaneous coordinate transformations preserve the Minkowski interval, demonstrating consistency with the invariance condition of Lorentz transformations.

2.2.2 The Lorentz Factor as a Coordinate Gradient

Having established the relationship between the additional coordinate and the Lorentz transformations, we can rearrange the spatial derivative to express the Lorentz factor directly in terms of $\partial_x y$.

As shown in equation 65,

$$
\partial_x y = \frac{\gamma v}{c} = \beta\gamma
\tag{21}
$$

Squaring both sides gives

$$
(\partial_x y)^2 = \frac{\gamma^2 v^2}{c^2}
\tag{22}
$$

Using the definition of the Lorentz factor,

$$
\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}},
\tag{23}
$$

we can rewrite this expression as

$$
(\partial_x y)^2 = \frac{v^2}{c^2 - v^2}.
\tag{24}
$$

Rearranging and substituting back into the Lorentz factor gives

$$
\begin{aligned}
\gamma^2 &= 1 + \frac{v^2}{c^2 - v^2} \\
&= 1 + (\partial_x y)^2.
\end{aligned}
\tag{25}
$$

Consequently, we obtain the following expression for the Lorentz factor in terms of the additional coordinate:

$$\boxed{\gamma = \sqrt{1 + (\partial_x y)^2}} \tag{26}$$

This establishes a direct relationship between the Lorentz factor and the spatial gradient of the additional coordinate, allowing the relativistic factor to be expressed geometrically without explicitly introducing velocity.

2.3 Numerical Correlations

2.3.1 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, these generialized coordinate transformations and a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

Earth's Kinematic Velocity From Electromagnetic Parameters

We define $\delta(M, R)$ to be

$$\delta(M, R) = \frac{d}{dR} g = 2\frac{GM}{R^3} \tag{27}$$

If $\delta(M, R)$ is induced by the coordinate transformations described above, we should presume that this dilation is radial. We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega(M, R) = \sqrt[3]{\frac{d}{dR} g} = (\delta(M, R))^{1/3} \tag{28}$$

Investigating this quantity for Earth gives

$$\omega_\oplus = 0.014537\ldots \tag{29}$$

While this approximates 2α, it more closely produces

$$\omega_\oplus = \alpha(2 - \alpha) \pm 0.0253\% \tag{30}$$

Then by expanding equation 30

$$\alpha(2 - \alpha) = \sqrt[3]{\frac{d}{dR} g} \tag{31}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4 \left(\frac{e^6 \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10} \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \right) = 1.272 \times 10^{21} \tag{32}$$

We then take the radius independent magnitude, isolating the additional electromagnetic parameters

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3}+\frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5}-\frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4}-\frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{33}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 8.

$$\gamma_{\text{em}}=1+\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3}+\frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5}-\frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4}-\frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{34}$$

$$v_{(\gamma_{\text{em}})}=371,721\ \text{m/s} \tag{35}$$

This matches the 2018 Planck data[12] to within 0.514%, and is in rough agreement with general observation of the velocity necessary to induce the CMB dipole.

Solar ω

We expand equation 31 and define $\Gamma(\sigma)$ to produce

$$\alpha=\Gamma(\sigma)\left(\frac{d}{dR}g\right)^{1/3} \tag{36}$$

Where $\Gamma(\sigma)$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 2 to derive certain characteristics of σ.

Table 2 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

In equation 31 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{\oplus}(\sigma)\approx\Gamma(\alpha)=\frac{1}{2-\alpha} \tag{37}$$

We propose that $\Gamma(\sigma)$ acts inside of the determinant. We then take the inverse of the determinant of a diagonal 3×3 matrix

$$\Gamma_{\odot}(\sigma)\propto\sqrt[3]{\Gamma(\sigma)} \tag{38}$$

We then presume that σ is associated with spin. Due to the increased surface gravitational acceleration and the decreased rate of spin for the Sun, we approximate $\sigma=0$.

$$\Gamma_{\odot}(\sigma) \approx \frac{1}{\sqrt[3]{2}} \tag{39}$$

Or

$$\alpha = \Gamma_{\odot}(\sigma) \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{40}$$

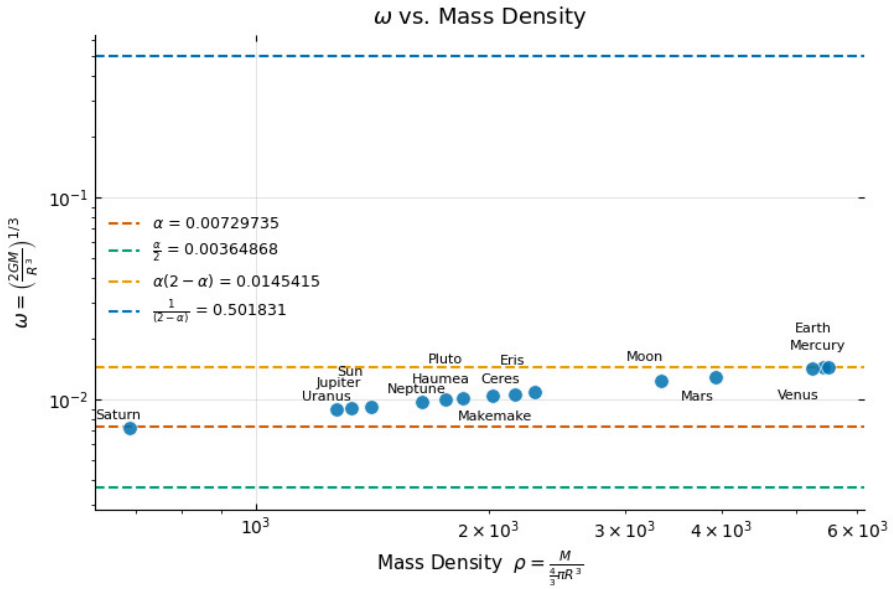


Fig. 2 The model quantity ω, identified with dy and constructed from the inverse determinant associated with volume change, plotted as a function of mass density.

We then find through observation that for the least mass dense body in the system, $\Gamma_{\hbar}(\sigma) \approx 1$.

This gives a set of 3 equations where the $\Gamma(\sigma)$ term cancels for the dominant body in the system.

$$\begin{aligned} g_{\oplus} &= R_{\oplus} \left(\frac{\alpha^3 (2-\alpha)^3}{2} \right) \tau && \pm 0.008\% \\ g_{\odot} &= R_{\odot} \alpha^3 \tau && \pm 1.426\% \\ g_{\hbar} &= \frac{1}{2} R_{\hbar} \alpha^3 \tau && \pm 1.134\% \end{aligned} \tag{41}$$

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha(2-\alpha)$ as shown in figure 2.

2.3.2 Gravitational Parameters

Due to algebraic symmetries, it's trivial to derive our kinematic velocity through gravitational parameters. What is of interest however, is that we derive what we propose is a more fundamental velocity first. It is only after integrating motion in a manner that is analogous to time that we produce our kinematic velocity. We call this first velocity our *cosmological velocity*, and propose a new geometric interpretation of existing data that may resolve the SNe/CMB velocity discrepancy.

Cosmological Velocity

As γ is applied as a scalar, we take the average of the integral of $\delta_{(M,R)}$ to be

$$\varphi\left(M,R\right)=\frac{1}{R}\int\delta\left(M,R\right)=1+G\frac{M}{R^3} \tag{42}$$

Substituting $\varphi\left(M,R\right)$ into equation 8 gives

$$\boxed{v_{\text{cosmo}}=c\sqrt{1-\frac{1}{\varphi^2(M,R)}}=525,493.17\text{ m/s}} \tag{43}$$

While this value aligns closely with the correlation carried out by Gordon et al.[13], there remains large uncertainties inherent in SNe observation[14, 15].

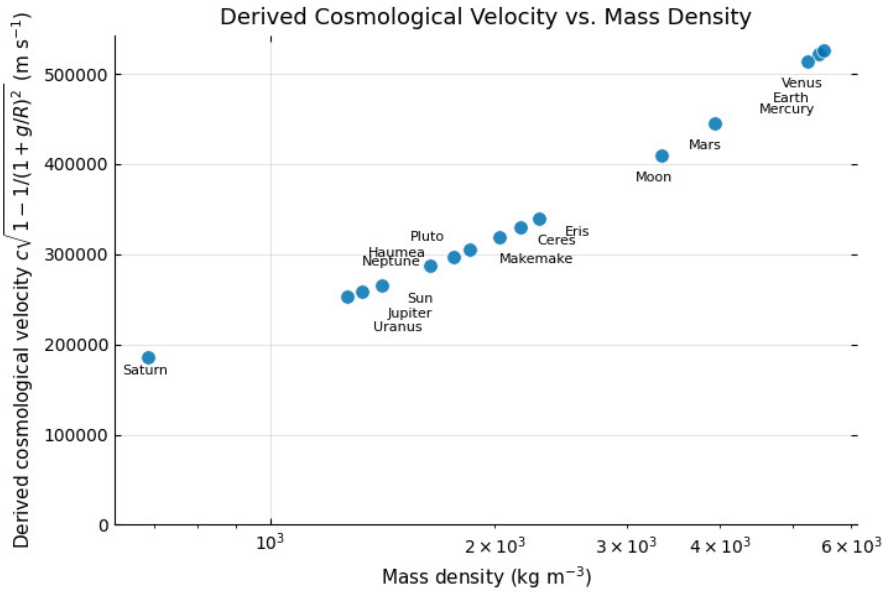


Fig. 3 Model-derived cosmological velocity plotted as a function of mass density.

As shown in figure 3, this function only holds for the theoretical upper bounds of mass density. We speculate that there may exist some yet unknown process in the

formation of solar systems, at least at our scale and location within the parent galaxy that leads to a natural function of mass density and velocity. While the details of this process have yet to be resolved, the coordinate transformations described here intrinsically link mass density to velocity as $\vec{g}$ becomes a function of v.

Kinematic Velocity

We apply the time-equivalent quantity in equation 6 to equation 42, making equation 42 time dependent. Integrating gives

$$\frac{1}{R}\int_0^R \int_0^x \delta\tau \; dR \; d\tau = 1 + \frac{1}{2}\frac{GM_\oplus}{R_\oplus^3}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{44}$$

We then substitute equation 44 into equation 8 where $d\tau = 1$

$$\boxed{v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,580 \text{ m/s}} \tag{45}$$

which matches the 2018 Plank data to within 0.47%.

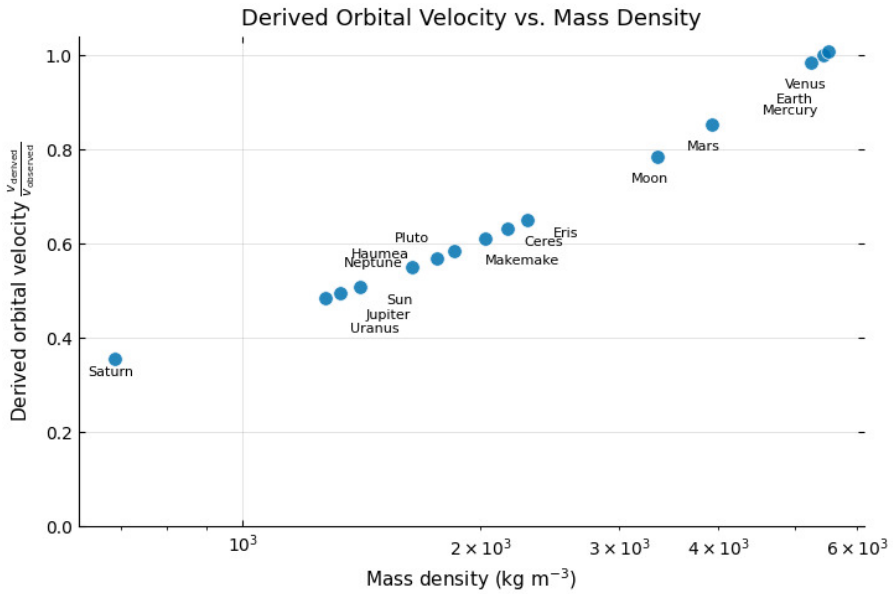


Fig. 4 Model-derived kinematic velocity as a ratio with the velocity found by Eilers et al. plotted as a function of mass density.

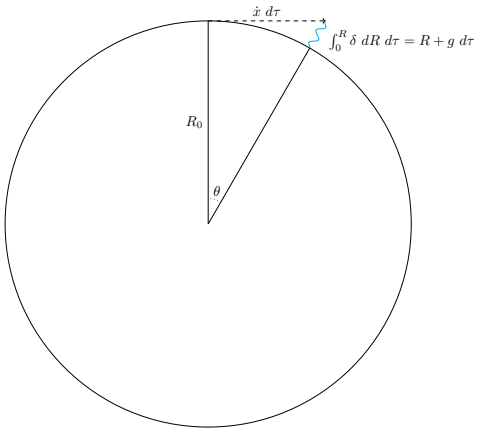


Fig. 5 Geometric construction used in the proposed stable-orbit derivation, showing R_0, the angle θ, the displacement $\dot{x}\,d\tau$, and the proposed radial increment $R + g\,d\tau$.

2.3.3 Orbital Velocity

Relativistic Stable Orbit Velocity

In the simplified single-body system considered here, the spatial dilation described above permits a trigonometric derivation of the stable-orbit velocity as an alternative to the conventional algebraic equality.

Figure 5 represents the geometric construction used in this derivation.

$$R_0^2 + v_{\text{orbit}}^2 d\tau^2 = \left(\int_0^{R_0} 2G \frac{M}{R^3} \right)^2 = (R_0 + g\ d\tau)^2 \tag{46}$$

Setting $\tau = 1$ gives

$$\boxed{v_{\text{orbit}}\,(M, R) = \frac{\sqrt{GM\,(GM + 2R^3)}}{R^2}} \tag{47}$$

For the comparison with the Milky Way, we adopt the enclosed Galactic baryonic mass reported by Cautun et al. [16]:

$$M_{\text{enc}} = M_{\text{bar}}(< R_0) = 4.44 \times 10^{10}\, M_\odot, \tag{48}$$

Here, $R_0 = 8.12\,\text{kpc}$ denotes the Sun's Galactocentric orbital radius[17, 18].

For comparison with observation, we use the solar-neighborhood circular velocity reported by Eilers et al. [19]. Using the adopted enclosed mass and solar orbital radius, the model gives the following orbital velocity:

$$v_{\text{orbit}}\,(M_{\text{enc}}, R_{\text{orbit}}) = 216,874\ \text{m/s} = v_{\text{eilers}} \pm 5.59\% \tag{49}$$

The model velocity equals the comparison velocity at $M_{\text{enc}} \approx 4.95 \times 10^{10} M_\odot$.

The adopted mass scale is also consistent with the characteristic stellar mass scale of approximately $5 \times 10^{10}\, M_\odot$ identified by Yu et al. [20] in their FIRE-2 simulations of Milky Way-mass galaxies. Yu et al. associate this scale with a transition in galactic

morphology from predominantly disc-dominated to increasingly spheroid-dominated systems.

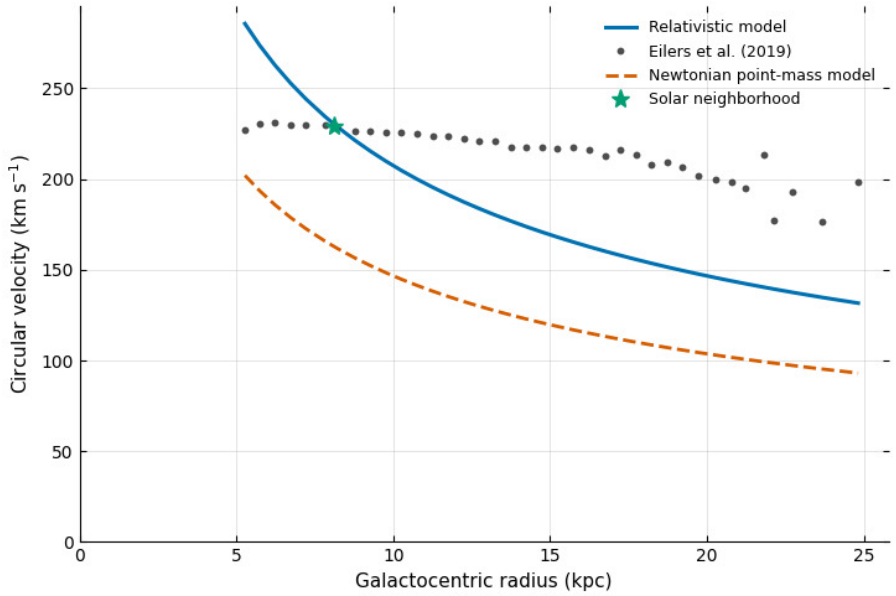


Fig. 6 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

It is important to note here that $v\left(M,R\right) \approx v_{\text{escape}}$ at $R = R_{\oplus}$ and $M = M_{\oplus}$, not a stable orbit velocity. This discrepency needs to be accounted for by Machian resistance in an n-body system. The associated equations remain to be evaluated, but geometric unity achieved elsewhere appears to indicate the viability of this geometry, such that the sum of this radiating force for all other bodies in the system may induce inertial resistance.

$v_{\text{orbit}}\left(M,R\right)$ should also only hold true where $M'\left(<R\right) \gg M'\left(>R\right)$, where M' is the sum of all masses as a function of radius, such as is the case on an outer arm of a spiral galaxy. This describes this function not as producing an orbital velocity as a function of radius, but as producing the upper limit of that function.

Kinematic Velocity From Triginometric Orbit Equations

We now investigate equation 49 by setting

$$v_{\text{orbit}} = \sqrt{2\frac{GM}{R}} \tag{50}$$

which is obtained by $E_{\text{kinetic}} = \int g\, dR$. Solving equation 46 for τ and substituting v_{orbit} gives

$$dt = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{51}$$

Inserting this quantity into equation 8 gives

$$v_{(|dt|)} = 371,580 \text{ m/s} \tag{52}$$

which matches the 2018 Plank data to within 0.47%.

2.4 Fifth-Dimensional Spatial Displacement and Gravitational Force

While most of the derivations to this point have relied on simplified single body solutions, we explore the connections to the KK 5-dimensional metric here. As the KK model imposes a cylindrical condition largely due to the lack of an observable 4th spatial dimension, we derive the magnitude of this 4th dimension from the coordinate transformations described above and apply no such condition.

2.4.1 Geometric Construction of the Fifth-Dimensional Displacement

We extend the ordinary three-dimensional displacement into the additional spatial coordinate y. Let dx denote the magnitude of the ordinary spatial displacement,

$$dx = \sqrt{dx_i dx^i}, \qquad i = 1, 2, 3. \tag{53}$$

For spatial velocity magnitude v and displacement parameter τ, as defined in the theoretical framework, the ordinary displacement is

$$dx = v\tau. \tag{54}$$

The proposed construction treats the extended displacement $dx^{(U)}$ as the hypotenuse of a right triangle whose perpendicular components are dx and dy as shown in figure 1. With γ denoting the Lorentz factor defined in Equation 8,

$$dx^{(U)} = v\tau\gamma. \tag{55}$$

Applying the Pythagorean theorem,

$$\left(dx^{(U)}\right)^2 = dx^2 + dy^2, \tag{56}$$

substitution gives

$$dy^2 = (v\tau\gamma)^2 - (v\tau)^2 \tag{57}$$

$$= v^2\tau^2\left(\gamma^2 - 1\right). \tag{58}$$

Taking the positive magnitude of the additional displacement yields

$$dy = v\tau\sqrt{\gamma^2 - 1} = \tau\frac{v^2}{\sqrt{c^2 - v^2}}. \tag{59}$$

The velocity form follows from the Lorentz-factor relation in Equation 8.

For constant v, differentiation with respect to τ gives the rate of displacement along y:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{60}$$

2.4.2 Incorporation into the Kaluza–Klein Metric

We represent the additional spatial coordinate through the metric component G_{44}. For a diagonal five-dimensional metric, the line element is

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{44}dy^2. \tag{61}$$

The corresponding metric tensor has the block structure

$$\boxed{G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & G_{44} \end{pmatrix}} \tag{62}$$

The component G_{44} sets the metric scaling of the additional coordinate. Its radial variation determines the coupling between motion along y and radial motion in the ordinary spatial coordinates.

For radial coordinate R, the Christoffel symbol associated with this coupling is

$$\Gamma^R_{44} = -\frac{1}{2}g^{RR}\frac{\partial G_{44}}{\partial R}. \tag{63}$$

For a normalized radial coordinate, $g^{RR} = 1$, so

$$\boxed{\Gamma^R_{44} = -\frac{1}{2}\frac{\partial G_{44}}{\partial R}} \tag{64}$$

2.4.3 Finding G_{44}

We take

$$\partial_x y = \frac{v}{\sqrt{c^2 - v^2}} \tag{65}$$

Or

$$dy = \tau\frac{v}{c^2 - v^2}dx \tag{66}$$

We then take the KK line element to be

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{44}dy^2 \tag{67}$$

and the $\mathbb{R}^4$ spatial separation to be

$$dS^2_{\text{spatial}} = dx^2 + dy^2 \tag{68}$$

where $dx^2 = dx_i dx^i$ for $i = 1, 2, 3$. Expanding this gives

$$dS^2_{\text{spatial}} = dx^2 \left(1 + \frac{v^2}{c^2 - v^2}\right) = dx^2 \gamma^2 \tag{69}$$

which matches the expected displacement magnitude of the Lorentz transformations.

Applying a flat metric to the line element defined in equation 67 gives

$$dS^2_U = dx^2 + G_{44} dy^2 \tag{70}$$

yielding

$$\boxed{G^{(U)}_{44} = 1} \tag{71}$$

for $dS^2_U = \gamma^2 dx^2$.

However solving equation 70 for G_{44} in the **S** coordinate system where $dS^2_S = d_i d^i$ gives

$$\boxed{G^{(S)}_{44} dy^2 = 0} \tag{72}$$

indicating that Newtonian mechanics may remain contained within a temporal surface, while the relativistic portion of displacement magnitude extends into the 4th spatial dimension, y. For this reason, we speculate that inertia has it's origins purely in the relativistic component of displacement; a consequence of a Machian-like resistance along the 4th spatial dimension.

2.4.4 Spatial curvature

Earlier we found $G^{(U)}_{44} = 1$ and $G^{(S)}_{44} dy^2 = 0$. Here we ask what fixes G_{44} within the radial geometry.

For the induced radial metric, only the product $G_{44}\,(dy/dR)^2$ enters,

$$h_{RR} = 1 + G_{44} \left(\frac{dy}{dR}\right)^2 . \tag{73}$$

The product alone cannot fix G_{44}, since a rescaling of y absorbs it. In this paper y is defined kinematically, $dy = v\tau\sqrt{\gamma^2 - 1}$, so $\partial_R y = \beta\gamma$ with βc the flow velocity, and G_{44} then has a definite value. Requiring the Schwarzschild radial component, $h_{RR} = 1/(1 - \beta^2) = 1 + \beta^2\gamma^2$ with $\beta^2 = r_s/R$, gives

$$\boxed{G_{44} = 1} \tag{74}$$

to all orders in r_s/R. To first order the radial component in areal coordinates is $1 + \gamma_{\text{PPN}}\, r_s/R$, so more generally

$$G_{44} = \gamma_{\text{PPN}} + \mathcal{O}\left(\frac{r_s}{R}\right). \tag{75}$$

Light deflection and Shapiro delay scale as $(1+\gamma_{\rm PPN})/2$, so G_{44} is measured by these tests. The Cassini radio-link measurement gives $\gamma_{\rm PPN}-1 = (2.1\pm2.3)\times10^{-5}$ [21], hence

$$G_{44} - 1 = (2.1 \pm 2.3) \times 10^{-5} \tag{76}$$

at 1σ, consistent with $G_{44} = 1$.

2.4.5 Field equation for G_{44}

Take the five-dimensional vacuum condition, imported here as an assumption, with a static spherical base metric $g_{\mu\nu}$ (Schwarzschild) and $G_{44} = \Phi(R)^2$ independent of y. The yy component of the Ricci tensor is $R_{yy} = -\Phi\,\Box\Phi$, so

$$\frac{1}{R^2}\frac{d}{dR}\left[R^2\left(1-\frac{r_s}{R}\right)\frac{d\Phi}{dR}\right] = 0 \quad \Rightarrow \quad \Phi = \Phi_\infty\left[1 + q\ln\left(1-\frac{r_s}{R}\right)\right]. \tag{77}$$

The constant q is a scalar charge. The profile diverges logarithmically at $R = r_s$, so regularity at the horizon requires $q = 0$ and $G_{44} = \Phi_\infty^2$ is constant. With the normalisation $G_{44}^{(U)} = 1$ this again gives $G_{44} = 1$.

2.4.6 Matter-Sourced Equation for the Fifth-Dimensional Metric

The vacuum solution for G_{44} obtained above does not determine how ordinary matter sources the fifth-dimensional geometry. To investigate this relationship, we replace the five-dimensional vacuum condition with the sourced Einstein equations,

$$G_{AB}^{(5)} = \kappa_5 T_{AB}^{(5)} \tag{78}$$

where κ_5 is the five-dimensional gravitational coupling and $T_{AB}^{(5)}$ is the five-dimensional stress-energy tensor. We use units in which $c = 1$ throughout this derivation.

We first consider the block-diagonal, pure-gradient sector of the metric,

$$dS^2 = g_{\mu\nu}(x)dx^\mu dx^\nu + \Phi(R)^2 dy^2 \tag{79}$$

where

$$G_{44} = \Phi^2 \tag{80}$$

The four-dimensional metric $g_{\mu\nu}$ is allowed to be curved, and Φ describes the radial variation of the fifth-dimensional metric component. For this calculation, we impose independence of the fifth coordinate,

$$\partial_y g_{\mu\nu} = 0, \qquad \partial_y \Phi = 0 \tag{81}$$

This restriction defines the y-independent sector of the theory.

Reduction of the five-dimensional field equations.

For the metric in equation (79), the relevant components of the five-dimensional Ricci tensor are

$$R^{(5)}_{\mu\nu} = R^{(4)}_{\mu\nu} - \frac{1}{\Phi}\nabla_\mu\nabla_\nu\Phi, \tag{82}$$

$$R^{(5)}_{44} = -\Phi\Box\Phi \tag{83}$$

Here, ∇_μ is the covariant derivative associated with $g_{\mu\nu}$, and

$$\Box\Phi = g^{\mu\nu}\nabla_\mu\nabla_\nu\Phi \tag{84}$$

is the four-dimensional covariant wave operator. The corresponding five-dimensional Ricci scalar is

$$R^{(5)} = R^{(4)} - \frac{2}{\Phi}\Box\Phi \tag{85}$$

Substituting these expressions into the five-dimensional Einstein tensor gives the four-dimensional components

$$G^{(5)}_{\mu\nu} = G^{(4)}_{\mu\nu} - \frac{1}{\Phi}\left(\nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Box\Phi\right) \tag{86}$$

while the fifth-dimensional component becomes

$$G^{(5)}_{44} = -\frac{1}{2}\Phi^2 R^{(4)} \tag{87}$$

The 44 field equation therefore determines the four-dimensional Ricci scalar in terms of the fifth-dimensional stress:

$$R^{(4)} = -2\kappa_5\frac{T^{(5)}_{44}}{\Phi^2} \tag{88}$$

Taking the four-dimensional trace of equation (86) gives

$$-R^{(4)} + \frac{3}{\Phi}\Box\Phi = \kappa_5 T^{(4)} \tag{89}$$

where

$$T^{(4)} = g^{\mu\nu}T^{(5)}_{\mu\nu} \tag{90}$$

is the trace of the four-dimensional projection of the five-dimensional stress-energy tensor. Substituting equation (88) into equation (89), we obtain

$$\boxed{\Box\Phi = \frac{\kappa_5\Phi}{3}\left(T^{(4)} - 2\frac{T^{(5)}_{44}}{\Phi^2}\right)} \tag{91}$$

Equation (91) is the sourced scalar field equation for the fifth-dimensional metric component in the y-independent sector. It relates the radial variation of G_{44} directly to the matter content of the system.

The matter source in a spherically symmetric system.

To identify the physical source, consider a static, spherically symmetric distribution of matter. Write the four-dimensional metric as

$$ds_4^2 = -N(R)^2 dt^2 + B(R)^2 dR^2 + R^2 d\Omega^2 \tag{92}$$

where $N(R)$ and $B(R)$ are the temporal and radial metric functions, respectively.

For an isotropic four-dimensional perfect fluid, let the four-dimensional stress-energy projection be

$$T_{\mu\nu}^{(5)} = (\epsilon + p) u_\mu u_\nu + p g_{\mu\nu} \tag{93}$$

where ϵ is the energy density, p is the ordinary isotropic pressure, and $u^\mu u_\mu = -1$. Its trace is

$$T^{(4)} = -\epsilon + 3p \tag{94}$$

We define the effective pressure in the fifth direction by

$$T_{44}^{(5)} = \Phi^2 p_y \tag{95}$$

where p_y is the stress associated with the additional spatial direction. Substitution into equation (91) gives

$$\boxed{\Box\Phi = -\frac{\kappa_5 \Phi}{3} (\epsilon - 3p + 2p_y)} \tag{96}$$

The fifth-dimensional scalar is therefore sourced by a particular combination of energy density and stresses. In the pressureless limit, with $p = p_y = 0$, the source reduces to the energy density.

For a static radial scalar, the covariant wave operator is

$$\Box\Phi = \frac{1}{NBR^2} \frac{d}{dR} \left(\frac{NR^2}{B} \frac{d\Phi}{dR} \right) \tag{97}$$

Consequently, the radial field equation becomes

$$\boxed{\frac{d}{dR} \left(\frac{NR^2}{B} \Phi' \right) = -\frac{\kappa_5}{3} NBR^2 \Phi \, (\epsilon - 3p + 2p_y)} \tag{98}$$

This equation determines the radial distribution of the fifth-dimensional metric directly from the matter distribution and the four-dimensional geometry.

Interior and exterior solutions in the weak-field limit.

An illustrative solution can be obtained for a spherical body of radius $R_\star$ with constant effective source

$$\sigma(R) \equiv \epsilon - 3p + 2p_y = \begin{cases} \sigma_0, & R \le R_\star, \\ 0, & R > R_\star. \end{cases} \tag{99}$$

In the weak-field limit, take

$$N \simeq 1, \qquad B \simeq 1, \qquad \Phi \simeq \Phi_\infty \tag{100}$$

on the right-hand side of equation (96). The scalar equation then reduces to

$$\frac{1}{R^2}\frac{d}{dR}\left(R^2\Phi'\right) = -\frac{\kappa_5\Phi_\infty}{3}\sigma. \tag{101}$$

Inside the body, regularity at the origin requires the integration constant multiplying R^{-2} to vanish. Integrating equation (101) gives

$$\Phi_{\rm in}(R) = C_0 - \frac{\kappa_5\Phi_\infty\sigma_0}{18}R^2. \tag{102}$$

Outside the body, the source vanishes, and the equation becomes

$$\frac{d}{dR}(R^2\Phi') = 0 \tag{103}$$

Its asymptotically normalized solution is

$$\Phi_{\rm out}(R) = \Phi_\infty + \frac{Q_\Phi}{R} \tag{104}$$

where Q_Φ is the scalar charge of the source.
Requiring continuity of Φ and its radial derivative at the surface gives

$$Q_\Phi = \frac{\kappa_5\Phi_\infty\sigma_0 R_\star^3}{9} \tag{105}$$

and

$$C_0 = \Phi_\infty + \frac{\kappa_5\Phi_\infty\sigma_0 R_\star^2}{6} \tag{106}$$

The matched solution is therefore

$$\boxed{\Phi(R) = \begin{cases} \Phi_\infty + \dfrac{\kappa_5\Phi_\infty\sigma_0}{18}\left(3R_\star^2 - R^2\right), & R \le R_\star, \\ \Phi_\infty + \dfrac{\kappa_5\Phi_\infty\sigma_0 R_\star^3}{9R}, & R > R_\star. \end{cases}} \tag{107}$$

The exterior solution demonstrates that a matter distribution can generate a nonconstant fifth-dimensional metric even when the exterior region itself is vacuum. The scalar charge is determined by the integrated interior source rather than by an independently prescribed exterior integration constant.

Interpretation and limitations.
Since $G_{44} = \Phi^2$, the exterior solution implies

$$G_{44}(R) = \Phi_\infty^2\left(1 + \frac{\kappa_5\sigma_0 R_\star^3}{9R}\right)^2 \tag{108}$$

To first order in the weak-field scalar perturbation,

$$G_{44}(R) \simeq \Phi_\infty^2 \left(1 + \frac{2\kappa_5 \sigma_0 R_\star^3}{9R}\right) \tag{109}$$

This result provides a direct matter-to-geometry relation for the fifth-dimensional metric component. It also identifies an important distinction between the present sourced solution and the vacuum solution derived previously: a nonzero exterior scalar profile is possible because its scalar charge is supplied by matter in the interior.

The calculation does not, by itself, determine the temporal metric function $N(R)$. That function must be obtained from the remaining components of the five-dimensional Einstein equations, which couple the four-dimensional curvature to the scalar field and the matter distribution. Consequently, the scalar profile derived here is a necessary part of the higher-dimensional dynamics, but it is not yet a complete derivation of the gravitational field or the orbital equations.

Finally, the derivation assumes independence of the fifth coordinate. The proposed construction treats the additional coordinate as an unwrapped spatial direction, and may therefore require dependence on y. Relaxing equation (81) introduces additional derivatives into the five-dimensional curvature and changes the scalar field equation. The present solution should thus be understood as the sourced, y-independent limit against which a more general unwrapped five-dimensional solution can be compared.

2.5 The Four-Potential

2.5.1 Restoring the Off-Diagonal Terms

The block-diagonal form of G_{AB} above sets $G_{\mu 4} = 0$. In Kaluza–Klein geometry the off-diagonal components are the four-potential. We restore them by writing the line element as

$$dS^2 = g_{\mu\nu}\, dx^\mu dx^\nu + G_{44}\,(dy + \mathcal{A}_\mu dx^\mu)^2 \tag{110}$$

so that

$$G_{AB} = \begin{pmatrix} g_{\mu\nu} + G_{44}\mathcal{A}_\mu\mathcal{A}_\nu & G_{44}\mathcal{A}_\mu \\ G_{44}\mathcal{A}_\nu & G_{44} \end{pmatrix}, \qquad G^{\mu\nu} = g^{\mu\nu}, \quad G^{\mu 4} = -\mathcal{A}^\mu, \quad G^{44} = \frac{1}{G_{44}} + \mathcal{A}_\mu\mathcal{A}^\mu \tag{111}$$

The line element is invariant under $y \to y + \Lambda(x)$, $\mathcal{A}_\mu \to \mathcal{A}_\mu - \partial_\mu \Lambda$. Setting $\mathcal{A}_\mu = 0$ recovers equation 62.

2.5.2 The Potential Generated by y

The coordinate transformations of this paper supply a natural candidate,

$$\boxed{\mathcal{A}_\mu \equiv \partial_\mu y} \tag{112}$$

evaluated in the rest frame of the source, where $y = y(x)$ is static. Equation 65 then gives

$$\mathcal{A}_0 = 0, \qquad \mathcal{A}_i = \beta\gamma\,\hat{n}_i \tag{113}$$

The quantity $\partial_\tau y = v\,\partial_x y$ used earlier is the rate of change of y along the worldline of the moving observer. It is the directional derivative $u^\mu \mathcal{A}_\mu$, not a component $\mathcal{A}_0$ of the field gradient. Three identities follow:

$$\gamma^2 = 1 + \mathcal{A}_i \mathcal{A}^i \tag{114}$$

$$\eta^{\mu\nu}\mathcal{A}_\mu\mathcal{A}_\nu = \beta^2\gamma^2 = \gamma^2 - 1, \qquad \mathcal{A}_\mu w^\mu = 0 \tag{115}$$

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu = 0 \tag{116}$$

where w^μ is the four-velocity of an observer at rest with respect to the source. Equation 114 is equation 26 restated, and equation 115 says the potential is a spacelike covector orthogonal to w^μ. Equation 116 holds because $\mathcal{A}_\mu$ is a pure gradient: in this sector the potential is pure gauge and carries no field strength.

2.5.3 Recovering the Induced Geometry

Transform to the frame in which the fibre coordinate is constant on a temporal surface, $y' = y - y(x)$, so that $dy' = 0$ on $\mathbf{S}$ and $\mathcal{A}_\mu = \partial_\mu y$. Equation 110 then restricts to the surface as

$$h_{\mu\nu} = g_{\mu\nu} + G_{44}\,\mathcal{A}_\mu\mathcal{A}_\nu \tag{117}$$

For $g_{ij} = \delta_{ij}$ and $G_{44} = 1$ this gives $h_{ij} = \delta_{ij} + \partial_i y\,\partial_j y$. Along a single direction it reproduces $dS^2 = \gamma^2 dx^2$, and for a radial profile $y(R)$ it reproduces

$$A(R) = h_{RR} = 1 + \left(\frac{dy}{dR}\right)^2 \tag{118}$$

the induced radial component of the embedding section. The construction developed so far is therefore the pure-gauge sector $\mathcal{F}_{\mu\nu} = 0$ of the full five-dimensional metric.

Remark. The orbit equations in this paper are geodesics of the intrinsic metric $h_{\mu\nu}$ on $\mathbf{S}$, that is, motion constrained to the surface. They are not geodesics of the five-dimensional metric. In conventional Kaluza–Klein theory a free particle conserves $p_y = G_{44}\,(\dot{y} + \mathcal{A}_\mu \dot{x}^\mu)$, which plays the role of charge, and $\mathcal{F}_{\mu\nu}$ acts as a force.

2.5.4 Curvature of the Surface and the Source

A hypersurface $y(x)$ of flat $\mathbb{R}^4$ has extrinsic curvature $K_{ij} = \partial_i\partial_j y/\sqrt{1 + \mathcal{A}_k\mathcal{A}^k}$, and its intrinsic scalar curvature follows from the Gauss equation,

$${}^{(3)}R = K^2 - K_{ij}K^{ij}. \tag{119}$$

For a spherically symmetric profile $y(R)$, define $\beta^2 \equiv \mathcal{A}_R^2/(1+\mathcal{A}_R^2) = 1-\gamma^{-2}$. Then

$${}^{(3)}R = \frac{2}{R^2}\frac{d}{dR}\left(R\,\beta^2\right). \tag{120}$$

Imposing the time-symmetric Hamiltonian constraint of general relativity, ${}^{(3)}R = 16\pi G\rho/c^2$, and integrating from a regular centre gives

$$\boxed{\beta^2(R) = \frac{2G\,m(R)}{c^2 R}, \qquad m(R) = \int_0^R 4\pi\rho R'^2\,dR'} \tag{121}$$

so βc is the escape velocity from the enclosed mass. The constraint is imported from general relativity; what the embedding adds is its expression through the Hessian of y, which ties mass density to the second derivative of the fourth coordinate. Three cases follow:

- Exterior, $m = M$: $\beta^2 = r_s/R$ and $y = \pm 2\sqrt{r_s(R - r_s)} + C$, the paraboloid found above.
- Uniform density: $\beta^2 = R^2/L^2$ with $L^2 = 3c^2/(8\pi G\rho)$, so $y = \pm\sqrt{L^2 - R^2} + C$, a spherical cap of a three-sphere of radius L. This is the spatial geometry of the interior Schwarzschild solution.
- Critical density: $\rho = 3H^2/(8\pi G)$ gives $L = c/H$ and $\beta = HR/c$, the Hubble flow. The Schwarzschild radius of a critical-density ball of radius c/H equals c/H.

Gravitational binding and cosmic expansion are then the same relation $\beta^2 = 2Gm/(c^2R)$, applied to different mass distributions. The sign of the associated flow differs: infall for a bound mass, outflow for the Hubble flow.

2.5.5 The Temporal Relation

Because $\mathcal{A}_0 = 0$ in the source frame, $h_{00} = g_{00}$, the embedding alone produces no gravitational redshift. This is expected, since the Schwarzschild spacetime is a class-two manifold, with a flat embedding known only in six dimensions [22], and cannot be embedded as a hypersurface of five-dimensional flat space. The temporal function must therefore be supplied separately. In terms of the potential, the simplest closure is

$$\boxed{N(R) = \frac{1}{\gamma} = (1 + \mathcal{A}_\mu \mathcal{A}^\mu)^{-1/2}} \tag{122}$$

which is equivalent to $N^2 = 1/A(R)$ and to the statement that the time-dilation factor is the Lorentz factor of the local escape velocity. It is a postulate, not a derivation. With it,

$$\vec{g} = \frac{c^2}{2}\nabla\beta^2, \qquad \delta(M, R) = \frac{c^2}{2}\frac{\partial^2\beta^2}{\partial R^2} = \frac{2GM}{R^3}, \tag{123}$$

which reproduces $\delta(M, R)$ from the equivalence-principle section as a derivative of a dimensionless potential. The same closure gives $v^2_{\text{local}} = GM/(R - r_s)$ in the exterior region, which is $\beta^2 c^2/2$ to leading order. Inside extended matter equation 122 is not the general-relativistic relation, so it should be applied to exterior regions only. The interior and time-dependent cases are treated in section 2.9.

2.5.6 Non-Gradient Component

The general potential is

$$\mathcal{A}_\mu = \partial_\mu y + a_\mu, \qquad \mathcal{F}_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu. \tag{124}$$

All results above correspond to $a_\mu = 0$. A non-zero a_μ is the only way for the five-dimensional geometry to carry a field strength, and $\mathcal{F}_{\mu\nu}$ acts on a body only through its fifth momentum p_y. We speculate that the spin dependence $\Gamma(\sigma)$ enters here as a curl component analogous to a gravitomagnetic potential, and possibly the electromagnetic parameters of section 2.3. Neither connection is derived here.

2.6 Radial Embedding and Relativistic Orbital Motion

Consider a point mass M located at the origin of a spherically symmetric spatial geometry. We assume that the radial coordinate R is associated with a corresponding position along an additional spatial coordinate y, such that the embedded hypersurface is described by

$$y = y(R; M) \tag{125}$$

Consequently, the differential displacement along the additional coordinate is

$$dy = \frac{dy}{dR}\, dR \tag{126}$$

We begin with a flat four-dimensional ambient spatial metric, restricted to the equatorial plane:

$$d\ell^2 = dR^2 + R^2 d\phi^2 + dy^2 \tag{127}$$

Substituting the radial embedding yields the induced spatial line element

$$d\ell^2_{\text{ind}} = dR^2 + R^2 d\phi^2 + \left(\frac{dy}{dR}\right)^2 dR^2 \tag{128}$$

$$= \left[1 + \left(\frac{dy}{dR}\right)^2\right] dR^2 + R^2 d\phi^2 \tag{129}$$

Defining the induced radial metric component as

$$A(R) = 1 + \left(\frac{dy}{dR}\right)^2 \tag{130}$$

the resulting metric becomes

$$\boxed{d\ell^2_{\text{ind}} = A(R) dR^2 + R^2 d\phi^2} \tag{131}$$

The radial metric component therefore contains the contribution from the gradient of the additional spatial coordinate. In particular,

$$\boxed{g^{\text{ind}}_{RR} = 1 + \left(\frac{dy}{dR}\right)^2} \tag{132}$$

2.6.1 The Orbit Equation

We introduce the reciprocal radial coordinate

$$u(\phi) = \frac{1}{R(\phi)}. \tag{133}$$

To describe geodesic motion on the induced spatial geometry, we introduce an affine parameter λ along the geodesic. The parameter λ is used solely to parameterize the spatial trajectory and is not an additional physical coordinate or the time-equivalent quantity τ introduced previously.

A dot denotes differentiation with respect to λ:

$$\dot{R} = \frac{dR}{d\lambda}, \qquad \dot{\phi} = \frac{d\phi}{d\lambda}. \tag{134}$$

For the induced spatial metric

$$d\ell^2_{\text{ind}} = A(R)dR^2 + R^2 d\phi^2, \tag{135}$$

the geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left[A(R)\dot{R}^2 + R^2\dot{\phi}^2\right]. \tag{136}$$

Since the metric does not explicitly depend on ϕ, the corresponding coordinate is cyclic. Its conjugate momentum is therefore conserved:

$$L = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = R^2\dot{\phi} = \text{constant}. \tag{137}$$

Using $u = 1/R$, this becomes

$$\dot{\phi} = Lu^2. \tag{138}$$

We now differentiate $R = 1/u$ with respect to the affine parameter λ:

$$\begin{aligned} \dot{R} &= \frac{dR}{d\phi}\dot{\phi} \\ &= -\frac{1}{u^2}\frac{du}{d\phi}Lu^2. \end{aligned} \tag{139}$$

Defining primes to denote differentiation with respect to ϕ, we obtain

$$\boxed{\dot{R} = -Lu'} \tag{140}$$

A second differentiation with respect to λ gives

$$\begin{aligned} \ddot{R} &= -L\frac{du'}{d\lambda} \\ &= -Lu''\dot{\phi} \\ &= -L^2u^2u''. \end{aligned} \tag{141}$$

Consequently,

$$\boxed{\ddot{R} = -L^2 u^2 u''} \tag{142}$$

Substituting these expressions into the radial geodesic equation produces

$$-L^2 u^2 u'' + \frac{A'}{2A} L^2 (u')^2 - \frac{L^2 u^3}{A} = 0. \tag{143}$$

Dividing by $-L^2 u^2$ gives the orbit equation of the induced spatial geometry:

$$\boxed{u'' + \frac{u}{A(1/u)} - \frac{A'(1/u)}{2A(1/u)u^2}(u')^2 = 0} \tag{144}$$

This equation describes spatial geodesics on the embedded hypersurface. It does not, by itself, describe relativistic particle orbits in spacetime, since no temporal metric component has yet been introduced.

Substituting these expressions into the radial geodesic equation produces

$$-L^2 u^2 u'' + \frac{A'}{2A} L^2 (u')^2 - \frac{L^2 u^3}{A} = 0 \tag{145}$$

Dividing by $-L^2 u^2$ gives the orbit equation of the induced spatial geometry:

$$\boxed{u'' + \frac{u}{A(1/u)} - \frac{A'(1/u)}{2A(1/u)u^2}(u')^2 = 0} \tag{146}$$

This equation describes spatial geodesics on the embedded hypersurface. It does not, by itself, describe relativistic particle orbits in spacetime, since no temporal metric component has yet been introduced.

2.6.2 The Schwarzschild Spatial Embedding

We can compare the induced radial metric with the spatial part of the Schwarzschild solution. At constant coordinate time and in the equatorial plane, the Schwarzschild spatial line element is

$$d\ell^2_{\text{Sch}} = \frac{dR^2}{1 - \dfrac{2GM}{c^2 R}} + R^2 d\phi^2 \tag{147}$$

Identifying the radial components of this metric with the induced radial metric gives

$$1 + \left(\frac{dy}{dR}\right)^2 = \frac{1}{1 - \dfrac{2GM}{c^2 R}} \tag{148}$$

Solving for the embedding gradient,

$$\left(\frac{dy}{dR}\right)^2 = \frac{1}{1 - \dfrac{2GM}{c^2 R}} - 1 \tag{149}$$

$$= \frac{\frac{2GM}{c^2R}}{1 - \frac{2GM}{c^2R}} \tag{150}$$

Introducing the Schwarzschild radius,

$$r_s = \frac{2GM}{c^2} \tag{151}$$

we obtain

$$\boxed{\frac{dy}{dR} = \pm\sqrt{\frac{r_s}{R - r_s}}} \tag{152}$$

Integration with respect to R gives

$$y(R) = \pm\sqrt{r_s} \int \frac{dR}{\sqrt{R - r_s}} \tag{153}$$

$$= \pm 2\sqrt{r_s(R - r_s)} + C \tag{154}$$

Consequently, the embedding profile is

$$\boxed{y(R) = \pm 2\sqrt{\frac{2GM}{c^2}\left(R - \frac{2GM}{c^2}\right)} + C} \tag{155}$$

This embedding reproduces the radial geometry of a Schwarzschild spatial slice for $R > r_s$. It demonstrates that the radial metric component can be represented geometrically through the gradient of an additional spatial coordinate.

However, reproducing the spatial geometry does not yet establish the complete relativistic dynamics. To investigate orbital motion, the temporal metric component must also be included.

2.6.3 The Full Spacetime Metric

Consider a static, spherically symmetric spacetime metric restricted to the equatorial plane:

$$ds^2 = -N(R)^2 c^2 dt^2 + A(R) dR^2 + R^2 d\phi^2 \tag{156}$$

Here, $A(R)$ is the radial metric component obtained from the spatial geometry, while $N(R)$ is the temporal metric function.

For the Schwarzschild solution,

$$A(R) = \frac{1}{1 - \frac{2GM}{c^2R}} \tag{157}$$

and

$$N(R)^2 = 1 - \frac{2GM}{c^2 R} \tag{158}$$

The relationship between these functions is therefore

$$\boxed{N(R)^2 = \frac{1}{A(R)}} \tag{159}$$

The spatial embedding determines $A(R)$ through the radial gradient of y. The corresponding temporal relation is an additional metric condition; it does not follow from the spatial embedding alone.

2.6.4 Conserved Quantities and the Radial Equation

For a massive particle, we take the proper time τ as the affine parameter. The geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left[-N(R)^2 c^2 \dot{t}^{\,2} + A(R)\dot{R}^{\,2} + R^2 \dot{\phi}^{\,2}\right] \tag{160}$$

Stationarity and spherical symmetry give two conserved quantities:

$$E = N(R)^2 c^2 \dot{t} \tag{161}$$

and

$$L = R^2 \dot{\phi} \tag{162}$$

The normalization of the four-velocity is

$$g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu = -c^2 \tag{163}$$

Substituting the metric and conserved quantities,

$$-N^2 c^2 \dot{t}^{\,2} + A\dot{R}^{\,2} + \frac{L^2}{R^2} = -c^2 \tag{164}$$

Since

$$\dot{t} = \frac{E}{N^2 c^2} \tag{165}$$

we obtain

$$-\frac{E^2}{N^2 c^2} + A\dot{R}^{\,2} + \frac{L^2}{R^2} = -c^2 \tag{166}$$

Solving for the radial velocity,

$$\boxed{A(R)\dot{R}^{\,2} = \frac{E^2}{N(R)^2 c^2} - c^2 - \frac{L^2}{R^2}} \tag{167}$$

This first integral describes the radial motion in terms of the two conserved quantities.

2.6.5 Derivation of the Relativistic Orbital Correction

We now substitute the Schwarzschild metric functions,

$$N(R)^2 = 1 - \frac{r_s}{R}, \qquad A(R) = \frac{1}{1 - \dfrac{r_s}{R}} \tag{168}$$

where $r_s = 2GM/c^2$.

The radial first integral becomes

$$\dot{R}^2 = \frac{E^2}{c^2} - \left(1 - \frac{r_s}{R}\right)\left(c^2 + \frac{L^2}{R^2}\right) \tag{169}$$

Introducing $u = 1/R$ and using

$$\dot{R} = -Lu', \tag{170}$$

we obtain

$$L^2(u')^2 = \frac{E^2}{c^2} - (1 - r_s u)(c^2 + L^2u^2) \tag{171}$$

Expanding the right-hand side,

$$L^2(u')^2 = \frac{E^2}{c^2} - c^2 - L^2u^2 + r_s c^2 u + r_s L^2 u^3 \tag{172}$$

Differentiating with respect to ϕ gives

$$2L^2u'u'' = -2L^2uu' + r_s c^2 u' + 3r_s L^2 u^2 u' \tag{173}$$

For a nonconstant radial trajectory, dividing by $2L^2u'$ yields

$$u'' = -u + \frac{r_s c^2}{2L^2} + \frac{3r_s}{2}u^2 \tag{174}$$

Rearranging,

$$u'' + u = \frac{r_s c^2}{2L^2} + \frac{3r_s}{2}u^2 \tag{175}$$

Finally, substituting $r_s = 2GM/c^2$, we arrive at

$$\boxed{\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2}u^2} \tag{176}$$

The first term on the right-hand side is the Newtonian contribution. The second term is the leading relativistic orbital correction.

In the Newtonian limit, $c \to \infty$, the relativistic contribution vanishes and the equation reduces to

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} \tag{177}$$

Its solution describes a conic-section orbit:

$$u(\phi) = \frac{GM}{L^2}\left[1 + e\cos(\phi - \phi_0)\right] \tag{178}$$

where e is the orbital eccentricity and ϕ_0 specifies the orientation of the orbit.

2.7 Circular Orbital Velocity and Stability

We now derive the circular orbital velocity directly from the spacetime metric. This provides an independent relationship between the metric components and the motion of a test particle, while also allowing us to establish the conditions under which circular orbits are stable.

We begin with the general static, spherically symmetric metric, restricted to the equatorial plane:

$$ds^2 = -N(R)^2c^2dt^2 + A(R)dR^2 + R^2d\phi^2 \tag{179}$$

Here, $N(R)$ represents the temporal metric function and $A(R)$ represents the radial metric component. Both are functions of the radial coordinate R.

2.7.1 Geodesic Equations

We parameterize the trajectory of a massive test particle by its proper time τ. The geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left[-N(R)^2c^2\dot{t}^{\,2} + A(R)\dot{R}^{\,2} + R^2\dot{\phi}^{\,2}\right] \tag{180}$$

where a dot denotes differentiation with respect to τ.

Because the metric is independent of both t and ϕ, these coordinates are cyclic. Their corresponding conjugate momenta are conserved. We define the conserved energy per unit mass and specific angular momentum as

$$E = N(R)^2c^2\dot{t}, \tag{181}$$

$$L = R^2\dot{\phi}. \tag{182}$$

Solving for the coordinate derivatives gives

$$\dot{t} = \frac{E}{N(R)^2c^2}, \tag{183}$$

$$\dot{\phi} = \frac{L}{R^2}. \tag{184}$$

We now apply the Euler–Lagrange equation to the radial coordinate:

$$\frac{d}{d\tau}\left(\frac{\partial\mathcal{L}}{\partial\dot{R}}\right) - \frac{\partial\mathcal{L}}{\partial R} = 0 \tag{185}$$

Evaluating the derivatives gives

$$\frac{d}{d\tau}\left(A(R)\dot{R}\right) + N(R)N'(R)c^2\dot{t}^{\,2} - \frac{1}{2}A'(R)\dot{R}^{\,2} - R\dot{\phi}^{\,2} = 0 \tag{186}$$

Expanding the total derivative and dividing by $A(R)$ yields the radial geodesic equation:

$$\ddot{R} + \frac{A'(R)}{2A(R)}\dot{R}^{\,2} + \frac{N(R)N'(R)c^2}{A(R)}\dot{t}^{\,2} - \frac{R}{A(R)}\dot{\phi}^{\,2} = 0 \tag{187}$$

2.7.2 Circular Orbital Velocity

For a circular orbit, the radial coordinate is constant. Therefore,

$$\dot{R} = 0, \qquad \ddot{R} = 0 \tag{188}$$

Substituting these conditions into the radial geodesic equation gives

$$\frac{N(R)N'(R)c^2}{A(R)}\dot{t}^{\,2} - \frac{R}{A(R)}\dot{\phi}^{\,2} = 0 \tag{189}$$

Multiplying by $A(R)$ and rearranging,

$$R\dot{\phi}^{\,2} = N(R)N'(R)c^2\dot{t}^{\,2} \tag{190}$$

The coordinate angular velocity is

$$\Omega = \frac{d\phi}{dt} = \frac{\dot{\phi}}{\dot{t}} \tag{191}$$

Dividing by $\dot{t}^{\,2}$, we obtain

$$\boxed{\Omega^2 = \frac{c^2 N(R)N'(R)}{R}} \tag{192}$$

To obtain the velocity measured by a local static observer, we must account for the relationship between coordinate time and proper time. At fixed radial coordinate, the metric becomes

$$ds^2 = -N(R)^2c^2dt^2 \tag{193}$$

Consequently, the observer's proper time satisfies

$$d\tau_{\text{static}} = N(R)dt \tag{194}$$

The local circumferential distance traversed by the particle is

$$d\ell = R\,d\phi \tag{195}$$

The locally measured orbital velocity is therefore

$$v_{\text{local}} = \frac{d\ell}{d\tau_{\text{static}}} = \frac{R}{N(R)}\frac{d\phi}{dt} \tag{196}$$

Squaring and substituting the expression for Ω^2 gives

$$v_{\text{local}}^2 = \frac{R^2}{N(R)^2}\Omega^2 \tag{197}$$

$$= \frac{R^2}{N(R)^2}\frac{c^2 N(R)N'(R)}{R}. \tag{198}$$

Thus, the locally measured circular orbital velocity is

$$\boxed{v_{\text{local}}^2 = c^2 R\frac{N'(R)}{N(R)}} \tag{199}$$

This result expresses circular orbital velocity directly in terms of the radial derivative of the temporal metric function.

2.7.3 Effective Potential and Orbital Stability

We next examine the conditions under which a circular orbit is stable. For a massive particle, the four-velocity normalization is

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -c^2 \tag{200}$$

Substituting the metric components gives

$$-N(R)^2c^2\dot{t}^2 + A(R)\dot{R}^2 + R^2\dot{\phi}^2 = -c^2 \tag{201}$$

Using the conserved quantities,

$$E = N(R)^2c^2\dot{t}, \tag{202}$$

$$L = R^2\dot{\phi}, \tag{203}$$

we obtain

$$-\frac{E^2}{N(R)^2c^2} + A(R)\dot{R}^2 + \frac{L^2}{R^2} = -c^2 \tag{204}$$

Multiplying by $N(R)^2$ and rearranging,

$$N(R)^2A(R)\dot{R}^2 + N(R)^2\left(c^2 + \frac{L^2}{R^2}\right) = \frac{E^2}{c^2} \tag{205}$$

We identify the effective potential as

$$\boxed{V_{\text{eff}}(R) = N(R)^2\left(c^2 + \frac{L^2}{R^2}\right)} \tag{206}$$

The radial equation can now be written as

$$N(R)^2A(R)\dot{R}^2 + V_{\text{eff}}(R) = \frac{E^2}{c^2} \tag{207}$$

A circular orbit occurs at an extremum of the effective potential:

$$\frac{dV_{\text{eff}}}{dR} = 0 \tag{208}$$

Differentiating while holding L constant gives

$$\frac{dV_{\text{eff}}}{dR} = 2N(R)N'(R)\left(c^2 + \frac{L^2}{R^2}\right) - \frac{2N(R)^2L^2}{R^3} \tag{209}$$

Setting this expression equal to zero and solving for L^2 yields

$$\boxed{L^2 = \frac{c^2R^3\left(N(R)^2\right)'}{2N(R)^2 - R\left(N(R)^2\right)'}} \tag{210}$$

A circular orbit is stable when the effective potential has a local minimum. The stability condition is therefore

$$\boxed{\frac{d^2V_{\text{eff}}}{dR^2} > 0} \tag{211}$$

If the second derivative is negative, the circular orbit is unstable. If it vanishes, the orbit is marginally stable and higher-order terms must be examined.

2.7.4 Schwarzschild Reference Case

We can verify these expressions using the Schwarzschild temporal metric function:

$$N(R)^2 = 1 - \frac{2GM}{c^2R} \tag{212}$$

Defining the Schwarzschild radius,

$$r_s = \frac{2GM}{c^2} \tag{213}$$

we can write

$$N(R)^2 = 1 - \frac{r_s}{R} \tag{214}$$

Differentiating,

$$\left(N(R)^2\right)' = \frac{r_s}{R^2} \tag{215}$$

Since

$$\frac{N'(R)}{N(R)} = \frac{\left(N(R)^2\right)'}{2N(R)^2} \tag{216}$$

the locally measured circular velocity becomes

$$v_{\text{local}}^2 = \frac{c^2R}{2}\frac{\left(N(R)^2\right)'}{N(R)^2} \tag{217}$$

Substituting the Schwarzschild function,

$$v_{\text{local}}^2 = \frac{c^2 R}{2} \frac{r_s/R^2}{1 - r_s/R} \tag{218}$$

$$= \frac{c^2 r_s}{2(R - r_s)}. \tag{219}$$

Therefore,

$$\boxed{v_{\text{local}}^2 = \frac{GM}{R - \frac{2GM}{c^2}}} \tag{220}$$

In the weak-field limit,

$$\frac{2GM}{c^2 R} \ll 1 \tag{221}$$

this expression reduces to the Newtonian circular orbital velocity:

$$v_{\text{local}} \approx \sqrt{\frac{GM}{R}} \tag{222}$$

We can also evaluate the angular momentum required for a Schwarzschild circular orbit. Substitution into the general expression gives

$$L^2 = \frac{c^2 R^3 (r_s/R^2)}{2(1 - r_s/R) - r_s/R} \tag{223}$$

$$= \frac{c^2 r_s R^2}{2R - 3r_s}. \tag{224}$$

Equivalently,

$$\boxed{L^2 = \frac{GMR^2}{R - \frac{3GM}{c^2}}} \tag{225}$$

Evaluating the second derivative of the effective potential at the circular-orbit condition gives

$$\boxed{V''_{\text{eff}}(R) = \frac{2c^2 r_s (R - 3r_s)}{R^3 (2R - 3r_s)}} \tag{226}$$

The existence of a timelike circular orbit requires

$$R > \frac{3}{2} r_s = \frac{3GM}{c^2} \tag{227}$$

The additional stability condition requires

$$R > 3r_s = \frac{6GM}{c^2} \tag{228}$$

Thus, the innermost stable circular orbit in Schwarzschild geometry is

$$\boxed{R_{\mathrm{ISCO}} = \frac{6GM}{c^2}} \tag{229}$$

2.8 Implications for the Higher-Dimensional Metric

The derivation establishes an important distinction between the radial and temporal components of the metric. The radial component determines the radial contribution to the line element, while the circular orbital velocity depends on the temporal metric function:

$$v_{\mathrm{local}}^2 = c^2 R \frac{N'(R)}{N(R)} \tag{230}$$

In the spatial embedding developed in this work, the radial metric component is related to the additional spatial coordinate through

$$A(R) = 1 + \left(\frac{dy}{dR}\right)^2 \tag{231}$$

This relationship determines the radial geometry of the embedded spatial surface. It does not independently determine the temporal metric function $N(R)$.

A complete derivation of circular orbital motion within the proposed higher-dimensional geometry therefore requires establishing $N(R)$ from the higher-dimensional construction. In particular, the relationship

$$N(R)^2 = \frac{1}{A(R)} \tag{232}$$

must be derived independently if it is to be used; it cannot be inferred solely from the radial embedding.

Once both metric components have been established, the general expressions derived above can be used to determine the predicted orbital velocity, angular momentum, and stability conditions.

2.9 Relation to the Painlevé–Gullstrand Form and to Cosmology

The identification $\beta^2 = 2Gm(R)/(c^2R)$ of equation 121 is not new in itself. This section places it in its established context, gives a dictionary between the quantities of this paper and their standard counterparts, and states what the construction adds and what it does not.

2.9.1 Schwarzschild in Painlevé–Gullstrand form

With $\beta^2 = r_s/R$ the Schwarzschild metric can be written [23–25]

$$ds^2 = -c^2 dT^2 + (dR + \beta c\, dT)^2 + R^2 d\Omega^2, \tag{233}$$

where T is the proper time of observers in radial free fall from rest at infinity. The surfaces $T = \text{const}$ are intrinsically flat, and βc is the Newtonian escape velocity, so space can be pictured as a flat background flowing inward at βc [26]. The rest space of an observer held at fixed R has radial metric component

$$h_{RR} = g_{RR} - \frac{g_{TR}^2}{g_{TT}} = 1 + \beta^2\gamma^2 = \gamma^2, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}, \tag{234}$$

so γ is the Lorentz factor between the infalling frame and the static observers, and the static proper length is $d\ell = \gamma\, dR$. This is the content of equations $dS^2 = dx^2 + dy^2 = \gamma^2 dx^2$ and $dy = \beta\gamma\, dx$ above, with x the coordinate R of the flat slices, y the Flamm embedding [27] of the static slice, and $\partial_R y = \beta\gamma$.

Along a free-fall worldline $dR/dT = -\beta c$, so $dy/dT = -c\beta^2\gamma$. With $v = \beta c$ its magnitude is $v^2/\sqrt{c^2 - v^2}$, which is equation 60, and its sign agrees with $c\, dt' = -\partial_\tau y$. The time-equivalent parameter τ therefore plays the role of the proper time T of the flow.

Temporal relation. For any $\beta(R)$, the change of time $dT = dt + \beta\, dR/\left[c(1-\beta^2)\right]$ takes equation 233 to

$$ds^2 = -(1-\beta^2)c^2 dt^2 + \frac{dR^2}{1-\beta^2} + R^2 d\Omega^2, \tag{235}$$

so $N^2 = 1/A = 1 - \beta^2$. The postulate of equation 122 is therefore equivalent to requiring that the frame moving at βc relative to the static observers has unit lapse and flat slices. For pressureless matter this is exact, because the dust frame is geodesic.

2.9.2 Hubble flow

For flat FLRW, $R = a(t)r$ gives $a\, dr = dR - HR\, dt$ and

$$ds^2 = -c^2 dt^2 + (dR - \beta_H c\, dt)^2 + R^2 d\Omega^2, \qquad \beta_H = \frac{HR}{c}, \tag{236}$$

the same form as equation 233 with the opposite sign of the shift. With $m = \frac{4\pi}{3}\rho R^3$ and the Friedmann equation $H^2 = 8\pi G\rho/3$, $\beta_H^2 = 2Gm/(c^2 R)$. The Hubble radius $R = c/H$ is where $\beta = 1$ and $2Gm/c^2 = R$, which is the apparent horizon of the flat FLRW universe [28, 29]. The Newtonian counterpart is the zero-energy condition of Milne and McCrea [30, 31]: the Hubble velocity equals the escape velocity from the enclosed mass.

2.9.3 The general relation

For spherical, pressureless, marginally bound dust (Lemaître–Tolman–Bondi [32–34] with zero energy function), $R'\, dr = dR - \dot R\, dt$ with $\dot R^2 = 2Gm/R$ gives equation 233 with $\beta^2 = 2Gm(R,T)/(c^2 R)$ and m the Misner–Sharp mass [35]. The generalised

Painlevé–Gullstrand form extends this to pressure [36]. Equation 121 is the same relation. Schwarzschild and flat FLRW are two cases of it:

Table 3 Cases of $\beta^2 = 2Gm/(c^2R)$ with unit lapse and flat slices

Configuration	$m(R)$	Flow	Static-slice embedding
Exterior of a mass	M	inward	paraboloid
Uniform density ball	$\frac{4\pi}{3}\rho R^3$	inward or outward	cap of a 3-sphere, $L^2 = 3c^2/8\pi G\rho$
Critical density (flat FLRW)	$\frac{4\pi}{3}\rho_c R^3$	outward	cap with $L = c/H$

Binding and expansion are then one relation with different mass distributions, and the sign of the flow distinguishes infall from Hubble expansion. The statement that gravitational acceleration and cosmic inflation are the same process, as observed from different frames, is made precise in this sense.

2.9.4 Scope

In the regimes above (vacuum or pressureless matter, spherical symmetry, flat-slice flow) the construction is equivalent to general relativity. The relation $\beta^2 = 2Gm/(c^2R)$ is taken from its Hamiltonian constraint, and $\dot{R}^2 = 2Gm/R$ is the Newtonian energy equation, so neither is derived here. What the construction adds is a reading of the flow as a coordinate gradient of a fourth spatial coordinate, with the Lorentz factor of the flow as the slope, together with the identification of the time-like parameter with the proper time of the flow. Predictions that differ from general relativity must come from sectors outside these regimes: a non-gradient potential a_μ, a varying G_{44}, or the relations of the equivalence-principle section.

2.9.5 Perihelion Precession

To determine the effect of the relativistic term, define the semilatus rectum

$$p = \frac{L^2}{GM} \tag{237}$$

The relativistic orbit equation becomes

$$u'' + u = \frac{1}{p} + \frac{3GM}{c^2}u^2 \tag{238}$$

In the weak-field limit,

$$\frac{GM}{pc^2} \ll 1 \tag{239}$$

we seek a first-order approximation to the Newtonian orbit. Substituting the leading-order form into the relativistic contribution gives the approximate solution

$$u(\phi) \simeq \frac{1}{p}\left[1 + e\cos\left(\left(1 - \frac{3GM}{pc^2}\right)\phi\right)\right] \tag{240}$$

The radial oscillation therefore does not repeat after an angular displacement of exactly 2π. Instead, the perihelion advances by

$$\Delta\phi = 2\pi \left[\frac{1}{1 - \dfrac{3GM}{pc^2}} - 1 \right] \tag{241}$$

To leading order,

$$\boxed{\Delta\phi = \frac{6\pi GM}{pc^2} = \frac{6\pi GM}{a(1-e^2)c^2}} \tag{242}$$

where a is the orbital semimajor axis and e is the eccentricity.
This is the leading-order Schwarzschild perihelion advance.

2.9.6 Geometric Interpretation

The derivation establishes a connection between the radial embedding and relativistic orbital geometry. The additional spatial coordinate produces an induced radial metric component,

$$A(R) = 1 + \left(\frac{dy}{dR} \right)^2, \tag{243}$$

which can reproduce the radial geometry of a Schwarzschild spatial slice. When the corresponding temporal metric function is also specified, the resulting spacetime geodesics produce the relativistic orbital correction.

The distinction between these two contributions is essential. The spatial embedding determines the radial geometry, whereas the temporal metric determines how the geometry relates to the evolution of a massive particle. The relativistic orbital correction follows from their combined effect, rather than from the spatial embedding alone.

Consequently, reproducing the Schwarzschild orbit equation through this construction requires establishing both the radial embedding and the corresponding temporal metric relation. Whether that temporal relation can be derived independently from the proposed fifth-dimensional geometry remains a separate question.

3 Discussion

The construction developed in this work investigates whether relativistic motion, gravitational geometry, and cosmological expansion can be described through a common geometric framework involving an additional spatial coordinate. Rather than introducing time as an independent geometric dimension at the outset, we investigate whether some of the relationships conventionally attributed to temporal evolution can be expressed through spatial displacement and changes in geometric scale.

Several results emerge from this approach. The coordinate transformations establish a relationship between the Lorentz factor and the gradient of an additional

spatial coordinate. The induced geometry reproduces the radial spatial metric of the Schwarzschild solution, while the corresponding spacetime construction recovers the relativistic orbital correction and the Schwarzschild circular-orbit stability conditions. Furthermore, the same dimensionless gravitational relation appears in the description of Schwarzschild geometry and homogeneous cosmological expansion.

These results provide a useful starting point for examining the relationship between the proposed geometry and existing gravitational theory. However, it is important to distinguish between geometric identities, results recovered from general relativity, numerical correlations, and predictions that remain specific to the proposed framework.

3.1 Geometric Interpretation of Relativistic Motion

A central feature of the proposed construction is the relationship between the Lorentz factor and the gradient of the additional spatial coordinate,

$$\gamma = \sqrt{1 + (\partial_x y)^2} \tag{244}$$

This expression allows the relativistic factor to be interpreted as a measure of the slope of an extended spatial geometry. The ordinary spatial displacement and the additional displacement combine to produce the extended displacement,

$$dx^{(U)} = \sqrt{dx^2 + dy^2} = \gamma dx \tag{245}$$

The significance of this relationship is that the additional coordinate provides a geometric representation of the relativistic contribution to displacement. In this interpretation, the Lorentz factor is no longer introduced exclusively as an algebraic correction to ordinary spatial motion, but can instead be associated with the geometry of an extended spatial surface.

This interpretation must nevertheless be distinguished from the standard Lorentz transformation. Recovering the Minkowski interval for the instantaneous transformations establishes an algebraic consistency condition, but does not independently establish the complete transformation of arbitrary spacetime events. In particular, the treatment of finite intervals, clock synchronization, and the relativity of simultaneity must be addressed before the proposed classical interpretation of simultaneity can be considered an alternative formulation of special relativity.

The additional coordinate also raises an important question concerning the physical interpretation of time. The parameter τ is introduced as a geometric quantity rather than as an independent temporal coordinate. However, its equivalence to conventional time must be established operationally. In particular, the dimensions of τ, the normalization of the reference length, and the relationship between derivatives with respect to τ and conventional time derivatives must be specified consistently.

Consequently, the present construction is best understood as a geometric reformulation under investigation, rather than an established replacement for the temporal structure of relativity.

3.2 The Additional Spatial Coordinate and Gravitational Geometry

The embedding construction provides a particularly useful connection between the proposed geometry and general relativity.

By representing the additional coordinate as a radial function $y(R)$, the induced spatial metric becomes

$$d\ell^2 = \left[1 + \left(\frac{dy}{dR}\right)^2\right] dR^2 + R^2 d\Omega^2 \tag{246}$$

For the Schwarzschild spatial geometry, the required embedding gradient is

$$\frac{dy}{dR} = \pm\sqrt{\frac{r_s}{R - r_s}} \tag{247}$$

where

$$r_s = \frac{2GM}{c^2} \tag{248}$$

This produces the familiar spatial embedding profile

$$y(R) = \pm 2\sqrt{r_s(R - r_s)} + C \tag{249}$$

The result demonstrates that the radial curvature of a Schwarzschild spatial slice can be represented through the gradient of an additional spatial coordinate.

An important distinction, however, is that reproducing a spatial metric does not independently establish the corresponding four-dimensional gravitational field. The spatial geometry and the temporal metric component must be considered together when describing the motion of physical particles.

This distinction becomes particularly important in the proposed five-dimensional construction. The metric component G_{44} controls the scaling of the additional coordinate, while its radial variation can contribute to the connection coefficients of the five-dimensional geometry. Nevertheless, the induced radial metric depends on the combination

$$h_{RR} = 1 + G_{44}\left(\frac{dy}{dR}\right)^2 \tag{250}$$

Consequently, the embedding alone does not uniquely determine G_{44} without an independently specified normalization of y.

Under the normalization adopted in this work, $G_{44} = 1$ reproduces the Schwarzschild spatial geometry. This is an important geometric result, but it should not be interpreted as demonstrating that the fifth-dimensional metric uniquely determines the gravitational field.

The distinction between an embedding coordinate and an independent physical degree of freedom will be particularly important in future developments of the model.

3.3 Relation to Kaluza–Klein Geometry

The five-dimensional metric provides a natural mathematical setting for investigating the additional spatial coordinate.

In the conventional Kaluza–Klein construction, the off-diagonal components of the higher-dimensional metric are associated with a four-dimensional vector potential. In the present framework, we identify a candidate potential through

$$\mathcal{A}_\mu = \partial_\mu y \tag{251}$$

This identification provides a direct relationship between the additional spatial coordinate and the four-dimensional metric components.

However, the potential obtained in this manner is a pure gradient. Its associated field strength therefore vanishes:

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu = 0 \tag{252}$$

This is an important restriction. Although the construction has the mathematical form of a Kaluza–Klein metric, the gradient sector alone does not reproduce a nonzero electromagnetic field strength.

A nontrivial field requires an additional component,

$$\mathcal{A}_\mu = \partial_\mu y + a_\mu \tag{253}$$

such that

$$\mathcal{F}_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu \tag{254}$$

The physical interpretation of this additional component remains open. In particular, the possible relationship between its antisymmetric derivatives, spin-dependent gravitational parameters, and electromagnetic interactions requires a field equation and independent physical justification.

The present results therefore establish a connection to the geometrical structure of Kaluza–Klein theory, but do not yet constitute a unified electromagnetic and gravitational theory.

3.4 Gravitational Binding and Cosmological Expansion

One of the more interesting aspects of the construction is the appearance of the same dimensionless relation in gravitationally bound systems and cosmological expansion:

$$\boxed{\beta^2(R) = \frac{2Gm(R)}{c^2 R}} \tag{255}$$

Here, $m(R)$ represents the enclosed mass and βc is the associated flow velocity.

For an exterior spherical mass distribution, $m(R) = M$, giving

$$\beta^2 = \frac{2GM}{c^2 R} \tag{256}$$

For a homogeneous cosmological distribution,

$$m(R) = \frac{4\pi}{3}\rho R^3 \tag{257}$$

and therefore

$$\beta^2 = \frac{8\pi G\rho R^2}{3c^2} \tag{258}$$

Applying the flat Friedmann relation,

$$H^2 = \frac{8\pi G\rho}{3} \tag{259}$$

we obtain

$$\beta = \frac{HR}{c} \tag{260}$$

Thus, the same geometric relationship describes the Schwarzschild radial flow and the Hubble flow, with opposite directions of motion.

This connection is closely related to the Painlevé–Gullstrand representation of Schwarzschild geometry and the corresponding areal-radius representation of a spatially flat FLRW universe. It is therefore not a new gravitational field equation. The underlying relation follows from established solutions and constraints of general relativity.

What the present construction adds is an interpretation of the associated flow through an additional spatial coordinate. In particular, the gradient of the embedding coordinate satisfies

$$\frac{dy}{dR} = \beta\gamma, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}} \tag{261}$$

The additional coordinate consequently represents the geometric displacement associated with the flow relative to the chosen spatial slicing.

This provides a common geometric description of gravitational binding and cosmological expansion without requiring the two situations to represent identical physical processes. Their different mass distributions and boundary conditions remain essential.

3.5 The Temporal Metric and Relativistic Orbital Motion

The relationship between the radial and temporal metric components is one of the most important remaining aspects of the proposed framework.

The induced spatial geometry determines

$$A(R) = 1 + \left(\frac{dy}{dR}\right)^2 \tag{262}$$

However, the motion of a massive particle depends on the complete spacetime metric,

$$ds^2 = -N(R)^2c^2dt^2 + A(R)dR^2 + R^2d\Omega^2 \tag{263}$$

The circular orbital velocity measured by a local static observer is

$$v_{\text{local}}^2 = c^2 R \frac{N'(R)}{N(R)} \tag{264}$$

This expression demonstrates that the circular orbital velocity depends directly on the temporal metric function and its radial gradient.

For Schwarzschild geometry, the relationship

$$N(R)^2 = \frac{1}{A(R)} = 1 - \frac{2GM}{c^2 R} \tag{265}$$

recovers the established relativistic circular-orbit velocity and the innermost stable circular orbit.

Similarly, the relativistic orbit equation becomes

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2}u^2 \tag{266}$$

Its weak-field solution produces the Schwarzschild perihelion advance,

$$\Delta\phi = \frac{6\pi GM}{a(1-e^2)c^2} \tag{267}$$

These results establish that the proposed spatial embedding can participate in a geometric construction that reproduces important features of Schwarzschild orbital dynamics when the corresponding temporal metric is supplied.

However, this distinction also identifies a fundamental limitation of the current model. The relationship between the temporal metric and the fifth-dimensional geometry has not yet been derived independently.

In particular, the condition

$$N(R)^2 = \frac{1}{A(R)} \tag{268}$$

is consistent with the Schwarzschild and appropriate Painlevé–Gullstrand constructions, but it cannot be inferred solely from the existence of the additional spatial coordinate.

Ideally, a more complete theory would establish this relationship from its own higher-dimensional dynamics, rather than importing it from an already known four-dimensional solution.

3.6 The Characteristic Velocities

The construction also produces two characteristic velocities associated with the gravitational parameters of the system.

The cosmological velocity is obtained from

$$v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2}} \tag{269}$$

whereas the kinematic velocity follows from the corresponding time-equivalent integration.

The numerical values obtained in this work are approximately

$$v_{\mathrm{cosmo}} = 525.5 \text{ km s}^{-1}, \tag{270}$$

$$v_{\mathrm{kinematic}} = 371.6 \text{ km s}^{-1} \tag{271}$$

Their ratio is approximately

$$\frac{v_{\mathrm{cosmo}}}{v_{\mathrm{kinematic}}} \approx \sqrt{2} \tag{272}$$

This relationship is particularly interesting because the same factor separates escape velocity from circular orbital velocity in the Newtonian limit.

The numerical proximity of the kinematic result to the velocity inferred from the CMB dipole provides motivation for further investigation, but the accuracy of the intersection would need to be improved with the development of a unified set of field equations. Likewise, the cosmological result motivates a comparison with velocity estimates obtained from Type Ia supernova observations, particularly those at large redshift.

Nevertheless, these comparisons involve physically distinct observables. The CMB dipole measures the motion of the observer relative to the CMB rest frame, whereas supernova-based bulk-flow estimates depend on the reconstruction of peculiar velocities from distance and redshift measurements.

Consequently, numerical agreement between the model velocities and these observations does not, by itself, establish that the quantities describe the same physical motion.

A meaningful comparison requires consistent reference frames, observer definitions, measurement uncertainties, and treatment of the underlying cosmological model. It must also account for the distinction between a local circular velocity, a gravitational escape velocity, and a cosmological peculiar velocity.

The factor of $\sqrt{2}$ provides a possible geometric connection between the characteristic velocities, but its physical interpretation requires further investigation.

3.7 Galactic Orbital Motion and the Baryonic Mass Distribution

The proposed orbital construction provides another motivation for further investigation.

Using the enclosed baryonic mass adopted from Cautun et al., the calculated solar-neighborhood velocity is approximately

$$v_{\mathrm{orbit}} = 216.9 \text{ km s}^{-1} \tag{273}$$

This is close to the observational value reported by Eilers et al., with a difference of approximately 5.6%.

The comparison is interesting because it uses an enclosed baryonic mass rather than introducing an additional dark matter component into the calculation.

However, agreement at a single galactocentric radius is not sufficient to establish that the model explains galactic rotation curves without dark matter.

A complete comparison requires the predicted velocity profile over a range of galactocentric radii, using the actual enclosed baryonic mass distribution rather than treating the Galaxy as a single point mass. The gravitational contributions of the stellar disk, bulge, gas, and other baryonic components must also be considered.

Furthermore, the orbital construction must be distinguished from the escape-velocity relation obtained from the radial embedding. As established in the preceding sections, the Schwarzschild circular velocity approaches the escape velocity divided by $\sqrt{2}$ in the weak-field limit.

The proposed model therefore requires an explicit treatment of the relationship between the geometric flow, circular motion, and the collective gravitational influence of an extended system.

The possibility that additional inertial effects arise from the distribution of matter throughout the system is worth investigating. However, a Machian interpretation of these effects requires a quantitative dynamical model before it can be used to modify the orbital predictions.

3.8 The Appearance of Electromagnetic Parameters

An additional observation concerns the repeated appearance of the fine structure constant in the gravitational approximations developed for Earth, the Sun, and Saturn.

The numerical relationships suggest the possibility that the gravitational parameters and the additional spatial coordinate share a common underlying function.

However, the present work does not derive a direct electromagnetic mechanism responsible for these relationships. Nor does it establish that the fine structure constant must appear in the gravitational field equations.

In particular, numerical agreement involving a dimensionful quantity requires careful treatment of units and normalization. A dimensionless fundamental constant cannot be identified with a dimensionful gravitational quantity without an additional scale or normalization.

The observed relationships should therefore be regarded as numerical correlations that motivate further theoretical investigation, rather than as evidence of an established electromagnetic origin of gravity.

A more complete treatment should establish whether these relationships follow from a dimensionally consistent higher-dimensional field equation, whether they depend on the properties of the individual source, and whether they produce predictions that can be independently tested.

3.9 Limitations and Future Directions

Several aspects of the proposed framework remain incomplete.

First, the higher-dimensional metric requires a complete dynamical treatment. Although the embedding provides a representation of the spatial geometry, the corresponding field equations for G_{44} and the four-potential have not yet been established as a complete system. The role of the non-gradient component and its possible relationship to spin and electromagnetic interactions also remains open.

Second, the temporal metric must be derived independently. The relationship $N^2 = 1/A$ provides the appropriate Schwarzschild result within the flow construction, but a general derivation must address extended matter distributions, pressure, and non-static sources.

Third, the orbital construction requires further development for extended and multiple-body systems. The relationship between the additional spatial displacement and inertial resistance must be expressed through a quantitative model before the proposed geometry can be used to make general predictions about orbital motion.

Finally, the cosmological and observational comparisons require systematic testing against independent datasets. In particular, the characteristic velocities should be evaluated using consistent reference frames and uncertainties.

These developments will determine whether the additional spatial coordinate represents only an alternative geometrical description of existing relativistic solutions or whether it can support new, independently testable physical predictions.

3.10 Concluding Remarks

The central proposition of this work is that gravitational geometry, relativistic displacement, and cosmological expansion may be investigated through a common description involving an additional spatial coordinate.

The Lorentz factor can be expressed through the gradient of this coordinate, the Schwarzschild spatial geometry can be represented as an induced hypersurface, and the same dimensionless gravitational relation appears in both bound gravitational systems and homogeneous cosmological expansion.

These connections provide a geometric framework in which several established relativistic relationships can be examined from a common perspective.

The results do not yet establish a complete alternative to general relativity, nor do they demonstrate that time is exclusively a spatial phenomenon. Rather, they identify a possible geometric relationship between the additional spatial displacement and the quantities conventionally associated with relativistic and gravitational evolution.

The next step is therefore not simply to reproduce additional known solutions, but to establish the higher-dimensional dynamics that determine the temporal metric, the gravitational field, and the behavior of matter within the proposed geometry.

If these relationships can be derived consistently and produce predictions distinguishable from existing gravitational theories, the additional spatial coordinate may provide a useful foundation for a broader geometrical description of gravitation and cosmology.

References

[1] Einstein, A.: On the electrodynamics of moving bodies. Annalen der Physik **17**(10), 891–921 (1905) https://doi.org/10.1002/andp.19053221004

[2] Einstein, A.: Kosmologische betrachtungen zur allgemeinen relativitätstheorie. Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin, 142–152 (1917)

[3] O'Raifeartaigh, C.: Albert einstein and the origins of modern cosmology. Physics Today (2017) https://doi.org/10.1063/PT.5.9085

[4] Nussbaumer, H.: Einstein's conversion from his static to an expanding universe. The European Physical Journal H **39**, 37–62 (2014) https://doi.org/10.1140/epjh/e2013-40037-6

[5] Kaluza, T.: Zum unitätsproblem der physik. Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin (Math. Phys.), 966–972 (1921)

[6] Klein, O.: Quantentheorie und fünfdimensionale relativitätstheorie. Zeitschrift für Physik **37**(12), 895–906 (1926) https://doi.org/10.1007/BF01397481

[7] Rossi, B., Hall, D.B.: Variation of the rate of decay of mesotrons with momentum. Physical Review **59**(3), 223–228 (1941) https://doi.org/10.1103/PhysRev.59.223

[8] Frisch, D.H., Smith, J.H.: Measurement of the relativistic time dilation using mu-mesons. American Journal of Physics **31**(5), 342–355 (1963) https://doi.org/10.1119/1.1969508

[9] Bailey, J., Borer, K., Combley, F., Drumm, H., Krienen, F., Lange, F., Picasso, E., Ruden, W., Farley, F.J.M., Field, J.H., Flegel, W., Hattersley, P.M.: Measurements of relativistic time dilatation for positive and negative muons in a circular orbit. Nature **268**, 301–305 (1977) https://doi.org/10.1038/268301a0

[10] Bailey, J., *et al.*: Final report on the cern muon storage ring including the anomalous magnetic moment and the electric dipole moment of the muon, and a direct test of relativistic time dilation. Nuclear Physics B **150**, 1–75 (1979) https://doi.org/10.1016/0550-3213(79)90292-X

[11] Botermann, B., Bing, D., Geppert, C., Gwinner, G., Hansch, T.W., Huber, G., Karpuk, S., Krieger, A., Kuhl, T., Nortershauser, W., Novotny, C., Reinhardt, S., Sanchez, R., Schwalm, D., Stohlker, T., Wolf, A., Saathoff, G.: Test of time dilation using stored li+ ions as clocks at relativistic speed. Physical Review Letters **113**(12), 120405 (2014) https://doi.org/10.1103/PhysRevLett.113.120405

[12] Planck Collaboration: Planck 2018 Results: Astro-ph Data Products. https://esa.int Accessed 2026-11-02

[13] Gordon, C., Land, K., Slosar, A.: Determining the motion of the solar system relative to the cosmic microwave background using type ia supernovae. Monthly Notices of the Royal Astronomical Society **387**(1), 371–376 (2008) https://doi.org/10.1111/j.1365-2966.2008.13239.x

[14] Brout, D., Scolnic, D., Popovic, B., *et al.*: The Pantheon+ analysis: Cosmological constraints. The Astrophysical Journal **938**(2), 110 (2022) https://doi.org/10.

3847/1538-4357/ac8e04

[15] Betoule, M., Kessler, R., Guy, J., *et al.*: Improved cosmological constraints from a joint analysis of the SDSS-II and SNLS supernova samples. Astronomy & Astrophysics **568**, 22 (2014) https://doi.org/10.1051/0004-6361/201423413

[16] Cautun, M., Benítez-Llambay, A., Deason, A.J., Frenk, C.S., Fattahi, A., Gómez, F.A., Grand, R.J.J., Oman, K.A., Navarro, J.F., Simpson, C.M.: The milky way total mass profile as inferred from gaia dr2. Monthly Notices of the Royal Astronomical Society **494**(3), 4291–4313 (2020) https://doi.org/10.1093/mnras/staa1017

[17] GRAVITY Collaboration: Detection of the gravitational redshift in the orbit of the star s2 near the galactic centre massive black hole. Astronomy & Astrophysics **615**, 15 (2018) https://doi.org/10.1051/0004-6361/201833718

[18] Bobylev, V.V., Bajkova, A.T.: A new estimate of the best value for the solar galactocentric distance. Astronomy Reports **65**(6), 498–506 (2021) https://doi.org/10.1134/S1063772921070015

[19] Eilers, A.-C., Hogg, D.W., Rix, H.-W., Ness, M.K.: The circular velocity curve of the milky way from 5 to 25 kpc. The Astrophysical Journal **871**(1), 120 (2019) https://doi.org/10.3847/1538-4357/aaf648

[20] Yu, S., Bullock, J.S., Gurvich, A.B., Hafen, Z., Stern, J., Boylan-Kolchin, M., Faucher-Giguère, C.-A., Wetzel, A., Hopkins, P.F., Moreno, J.: Born this way: thin disc, thick disc, and isotropic spheroid formation in fire-2 milky way-mass galaxy simulations. Monthly Notices of the Royal Astronomical Society **523**(4), 6220–6238 (2023) https://doi.org/10.1093/mnras/stad1806

[21] Bertotti, B., Iess, L., Tortora, P.: A test of general relativity using radio links with the Cassini spacecraft. Nature **425**(6956), 374–376 (2003) https://doi.org/10.1038/nature01997

[22] Kasner, E.: Finite representation of the solar gravitational field in flat space of six dimensions. American Journal of Mathematics **43**, 130–133 (1921)

[23] Painlevé, P.: La mécanique classique et la théorie de la relativité. Comptes Rendus de l'Académie des Sciences (Paris) **173**, 677–680 (1921)

[24] Gullstrand, A.: Allgemeine Lösung des statischen Einkörperproblems in der Einsteinschen Gravitationstheorie. Arkiv för Matematik, Astronomi och Fysik **16**(8), 1–15 (1922)

[25] Martel, K., Poisson, E.: Regular coordinate systems for Schwarzschild and other spherical spacetimes. American Journal of Physics **69**, 476 (2001) gr-qc/0001069

[26] Hamilton, A.J.S., Lisle, J.P.: The river model of black holes. American Journal of Physics **76**, 519–532 (2008) https://doi.org/10.1119/1.2830526

[27] Flamm, L.: Beiträge zur Einsteinschen Gravitationstheorie. Physikalische Zeitschrift **17**, 448–454 (1916)

[28] Faraoni, V.: Cosmological apparent and trapping horizons (2011) 1106.4427

[29] Faraoni, V., Vachon, G., Vanderwee, R., Jose, S.: Friedmann–Lemaître–Robertson–Walker cosmology through the lens of gravitoelectromagnetism (2022) 2201.05287

[30] Milne, E.A.: A Newtonian expanding universe. Quarterly Journal of Mathematics (Oxford) **5**, 64–72 (1934)

[31] McCrea, W.H., Milne, E.A.: Newtonian universes and the curvature of space. Quarterly Journal of Mathematics (Oxford) **5**, 73–80 (1934)

[32] Lemaître, G.: L'Univers en expansion. Annales de la Société Scientifique de Bruxelles A **53**, 51–85 (1933)

[33] Tolman, R.C.: Effect of inhomogeneity on cosmological models. Proceedings of the National Academy of Sciences **20**, 169–176 (1934)

[34] Bondi, H.: Spherically symmetrical models in general relativity. Monthly Notices of the Royal Astronomical Society **107**, 410–425 (1947)

[35] Misner, C.W., Sharp, D.H.: Relativistic equations for adiabatic, spherically symmetric gravitational collapse. Physical Review **136**, 571 (1964)

[36] Lasky, P.D., Lun, A.W.C.: Generalized Lemaître–Tolman–Bondi solutions with pressure. Physical Review D **74**, 084013 (2006)